

Solution Requirements

Date	01 July 2026
Group ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR):

S.No	Requirement Category	Requirement Description	Priority
1	User Input Management	Accept Life Expectancy, Expected Years of Education, Mean Years of Education, and GNI per Capita from users.	High
2	Input Validation	Validate user inputs before processing predictions.	High
3	External Interfaces	Connect datasets and machine learning libraries.	High
4	Prediction Processing	Generate HDI predictions using the trained Machine Learning model.	High
5	Result Visualization	Display predicted HDI score and development category.	Medium
6	Business Rules	Prediction based on health, education and income factors.	High
7	Data Handling	Manage dataset loading and model execution efficiently.	Medium
8	Model Management	Load and utilize the trained HDI prediction model (<code>HDI.pkl</code>).	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric
1	Performance & Speed	Generate HDI predictions quickly.	Few seconds response
2	Scalability	Support larger datasets and future enhancements.	Expandable system
3	Security & Data Privacy	Protect application data and processing logic.	Secure authentication
4	Reliability & Availability	Ensure consistent and stable operation.	High availability
5	Usability & Accessibility	Provide a simple and user-friendly interface	User friendly
6	Maintainability	Allow easy updates to dataset and ML model.	Modular structure

System Overview:

1. **A Comprehensive Measure of Well-Being** is a Machine Learning-based web application that predicts the Human Development Index (HDI).
2. The system accepts key development indicators such as:
 - Life Expectancy at Birth
 - Expected Years of Education
 - Mean Years of Education
 - Gross National Income (GNI) per Capita
3. Python, Flask, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn are used for development.
4. The trained Machine Learning model analyzes development indicators and predicts the HDI score and corresponding development category.
5. The system provides prediction results and supports analysis of factors affecting human development and well-being.